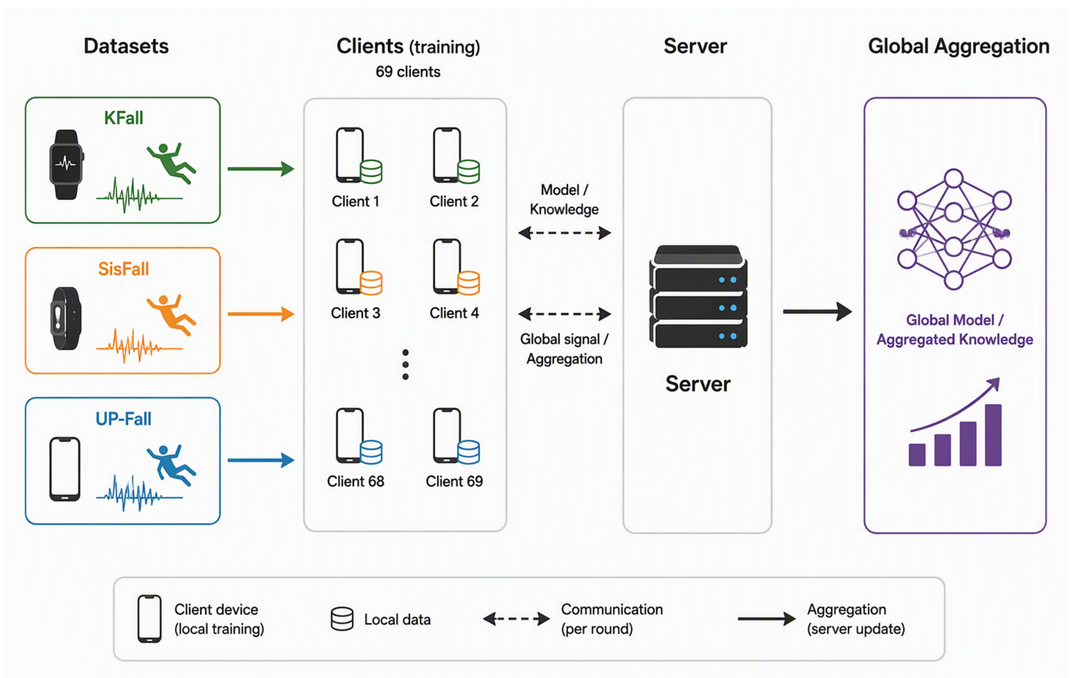
Revisiting Communication in Federated Learning: Parameter vs. Knowledge-Based Strategies

Leonor Ramos[1], Carlos Serrão[1] e Ana de Almeida[1,2]
1—Instituto Universitário de Lisboa (ISCTE-IUL), ISTAR, Lisboa, Portugal
2—University of Coimbra, CISUC/LASI, Coimbra, Portugal

Motivation & Research Goals

Federated Learning (FL) enables collaborative model training without sharing raw data, making it particularly suitable for privacy-sensitive healthcare applications. However, performance strongly depends on the communication mechanism used between clients and the server, especially under heterogeneous and non-IID data distributions. This work compares parameter-sharing and knowledge-based communication strategies for wearable fall detection, nvestigating their impact on predictive performance and communication efficiency.

Methods



Datasets: KFall, SisFall and UP-Fall
Experimental Setup: 69 Training Clients, 9 Development Clients and 9 Test Clients
Model : $32 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 1$

Strategy	Transferred Information
FedAvg	Model weights
KD + Weight Sharing	Model weights + distillation
Soft Labels	Prediction statistics
Class Prototypes	Class-level representations

Selected Results

Global vs Local Performance

Strategy	Global PR-AUC	Local PR-AUC
FedAvg	0.969	0.915
KD + Weight Sharing	0.957	0.914
Soft Labels	0.737	0.817
Class Prototypes	0.701	0.819

Communication Cost

Strategy	Values Exchanged
FedAvg	49,665
KD + Weight Sharing	49,665
Class Prototypes	131
Soft Labels	2

Main Finding
FedAvg achieved the highest global performance (PR-AUC = 0.969). However, knowledge-based communication strategies substantially reduced communication costs while maintaining competitive local performance. Soft-label communication exchanged only 2 values per round and achieved a local PR-AUC of 0.817.

Conclusion

FedAvg achieved the best global performance under heterogeneous non-IID data. Knowledge-based communication strategies achieved lower global performance but maintained competitive local performance while dramatically reducing communication costs. These results suggest that lightweight knowledge-sharing mechanisms constitute practical alternatives for communication-constrained federated healthcare applications.

Key References

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For the complete paper, supplementary material, and full bibliography, please scan the QR code.

Acknowledgements

This work was partially supported by national funds through Fundação para a Ciência e a Tecnologia, I.P. (FCT), ISTAR-Iscte Pluriannual Projects UID/04466/2025 (https://doi.org/10.54499/UID/04466/2025).